SIMPLIFICATION

SET 1

SQUARE ROOTS

1. **Simplify $\sqrt{900} + \sqrt{0.09} + \sqrt{0.000009}$ = ?**
   A. 30.27 C. 30.097
   B. 30.297 D. 30.197

2. $$\sqrt{\frac{0.009 \times 0.036 \times 0.016 \times 0.08}{0.002 \times 0.0008 \times 0.0002}}$$
   A. 34 C. 36
   B. 38 D. 39

3. $$\sqrt{1\frac{1}{4} \times \frac{64}{125} \times 1.44}$$
   A. $1\frac{1}{25}$ C. $\frac{23}{25}$
   B. $\frac{24}{25}$ D. $\frac{21}{25}$

4. **$2\sqrt{54} - 6\sqrt{\frac{2}{3}} - \sqrt{96}$ =**
   A. 0 B. 1 C. 2 D. $\sqrt{6}$

5. $\frac{\sqrt{24}+\sqrt{216}}{\sqrt{96}}$ =
   A. $\frac{2}{\sqrt{6}}$ C. $4\sqrt{6}$
   B. $2\sqrt{6}$ D. 2

6. $$(3+\sqrt{8}) + \frac{1}{3-\sqrt{8}} - (6+4\sqrt{2})$$
   A. 8 B. 1 C. $\sqrt{2}$ D. 0

7. $\sqrt{\frac{0.441}{0.625}}$ =
   A. 0.048 C. 0.48
   B. 0.84 D. 0.084

8. **The square root of** $\frac{0.342 \times 0.684}{0.000342 \times 0.000171}$
   A. 250 C. 2000
   B. 2500 D. 4000

9. **$\sqrt{0.00060516}$ =**
   A. 0.0246 C. 0.246
   B. 0.00246 D. 0.000246

10. **If 102^2 = 10404 ; find $\sqrt{104.04} + \sqrt{1.0404} + \sqrt{0.010404}$ =**
    A. 0.306 C. 11.122
    B. 0.0306 D. 11.322

11. **$\sqrt{0.00004761}$ =**
    A. 0.069 C. 0.00069
    B. 0.0069 D. 0.0609

12. **Given that $\sqrt{574.6}$ = 23.97 , $\sqrt{5746}$ =75.8; find $\sqrt{0.00005746}$ = ?**
    A. 0.002397 C. 0.007580
    B. 0.0002397 D. 0.00758

13. **Square root of $(7+3\sqrt{5})(7-3\sqrt{5})$ =**
    A. 4 C. $3\sqrt{5}$
    B. $\sqrt{5}$ D. 2

14. **The value of $\sqrt{400}$ + $\sqrt{0.0400}$ + $\sqrt{0.000004}$ =**
    A. 0.222 C. 20.202
    B. 20.22 D. 2.022

15. **If $\sqrt{3}$ = 1.7321, find $\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75}$ =**
    A. 8.661 C. 1.7321
    B. 4.331 D. -1.732

16. **If $\sqrt{4096}$ =64; find** $\sqrt{40.96} + \sqrt{0.4096} + \sqrt{0.004096} + \sqrt{0.00004096}$ =
    A. 7.09 C. 7.11
    B. 7.10 D. 7.12

17. $$\sqrt{8+\sqrt{57+\sqrt{38+\sqrt{108+\sqrt{169}}}}}$$
    A. 4 B. 6 C. 8 D. 10

18. **If 10.15^2 = 103.02225 , then**
    $$\sqrt{1.030225} + \sqrt{10302.25} =$$
    A. 1025.15 C. 103.515
    B. 102.515 D. 102.0515

19. **If $\sqrt{18225}$ = 135; then** $\sqrt{18225} + \sqrt{182.25} + \sqrt{1.8225} + \sqrt{0.018225}$
    A. 14.9985 C. 1499.85
    B. 149.985 D. 1.49985

20. $$\sqrt{21\frac{51}{169}} =$$
    A. $5\frac{8}{13}$ C. $4\frac{3}{13}$
    A. $4\frac{8}{13}$ D. $5\frac{5}{13}$

21. **If $(1101)^2$ = 1212201; find $\sqrt{121.2201}$ =**
    A. 110.01 C. 1.101
    B. 11.01 D. 11.001

22. $$\sqrt{\frac{0.064 \times 0.256 \times 15.625}{0.025 \times 0.625 \times 4.096}} =$$
    A. 2 C. 0.24
    B. 2.4 D. 4.2

23. **Find $\sqrt{19.36} + \sqrt{0.1936} + \sqrt{.001936} + \sqrt{0.00001936}$ =**
    A. 4.8484 C. 4.8884
    B. 4.8694 D. 4.8234

24. **Simplify**
    $$\sqrt{3\frac{33}{64}} \div \sqrt{9\frac{1}{7}} \times 2\sqrt{3\frac{1}{9}}$$
    A. $\frac{45}{256}$ C. $4\frac{3}{8}$
    B. $1\frac{17}{28}$ D. $2\frac{3}{16}$

25. **Simplify** $\frac{\sqrt{32}+\sqrt{48}}{\sqrt{8}+\sqrt{12}}$ =
    A. 3 C. 6
    B. 2 D. 4

26. **Simplify**
    $$\frac{3\sqrt{2}}{\sqrt{3}+\sqrt{6}} - \frac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}} + \frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}}$$
    A. $\sqrt{2}$ C. $\sqrt{3}-\sqrt{2}$
    B. $\frac{1}{\sqrt{2}}$ D. 0

27. $\sqrt{\frac{2+\sqrt{3}}{2}}$
    A. $\pm\frac{1}{\sqrt{2}}(\sqrt{3}+1)$
    B. $\pm\frac{1}{2}(\sqrt{3}-2)$
    C. $\pm\frac{1}{2}(\sqrt{3}-1)$
    D. None of these

SET 2

CUBE ROOTS

1. $\frac{\sqrt[3]{8}}{\sqrt{16}} \div \sqrt{\frac{100}{49}} \times \sqrt[3]{125}$ =
   A. 7 C. $\frac{7}{100}$
   B. 1 ¾ D. $\frac{4}{7}$

2. $\sqrt[3]{\frac{72.9}{0.4096}}$ =
   A. 0.5625 C.182
   B. 5.625 D. 13.6

3. **$(5.5)^3 - (4.5)^3$ =**
   A. 1 C. 74.25
   B. 75 D. 75.25

4. $\sqrt[3]{\frac{7}{875}}$ =
   A. $\frac{1}{3}$ C. $\frac{1}{4}$
   B. $\frac{1}{15}$ D. $\frac{1}{5}$

5. $\sqrt[3]{\frac{19}{513}}$ =
   A. $\frac{1}{9}$ C. $\frac{1}{\sqrt{27}}$
   B. $\frac{1}{3}$ D. $\frac{1}{\sqrt{3}}$

6. **If $\sqrt[3]{175616}$ = 56;**

$\sqrt[3]{175.616} + \sqrt[3]{0.175616} + \sqrt[3]{0.000175616} =$

A. 0.168 C. 6.216
B. 62.16 D. 6.116

7. $\sqrt[3]{\sqrt{0.000064}} =$

A. 0.0002 C. 0.02
B. 0.002 D. 0.2

8. $\sqrt[3]{15612 + \sqrt{154 + \sqrt{225}}}$

A. 15 C. 75
B. 25 D. 125

9. $\sqrt[3]{0.000125} =$

A. 0.5 C. 0.05
B. 0.15 D. 0.005

10. $\sqrt[3]{1000} + \sqrt[3]{0.008} - \sqrt[3]{0.125} =$

A. 9.7 C. 9.997
B. 9.97 D. 9.9997

11. $\sqrt[3]{1 - \frac{127}{343}} =$

A. $\frac{5}{9}$ B. $\frac{4}{7}$
A. $1 - \frac{1}{7}$ D. $1 - \frac{2}{7}$

12. **If $\sqrt[3]{3^n} = 27$, find n?**

A. 9 B. 1
C. 6 D. 3

13. $\sqrt{\sqrt[3]{0.000729}} =$

A. 0.9 C. 0.03
B. 0.3 D. 0.09

14. $\sqrt[3]{4\frac{12}{125}} =$

A. 1.4 C. 1.8
B. 1.6 D. 2.4

15. **If $X = \sqrt{3} + \sqrt{2}$, find $x^3 - \frac{1}{x^3}$?**

A. $10\sqrt{2}$ C. $22\sqrt{2}$
B. $14\sqrt{2}$ D. $8\sqrt{2}$

16. **If $\sqrt[3]{79507} = 43$,** $\sqrt[3]{79.507} + \sqrt[3]{0.079507} + \sqrt[3]{0.000079507} =$

A. 0.4773 C. 47.73
B. 477.3 D. 4.773

17. **When simplified, the product of :** $(2 - \frac{1}{3})(2 - \frac{3}{5})(2 - \frac{5}{7})$ ....$(2 - \frac{997}{999})$

A. $\frac{5}{999}$ C. $\frac{1001}{999}$
B. $\frac{5}{3}$ D. $\frac{1001}{3}$

18. **The sum of the cubes of two numbers is 793. The sum of the numbers is 13. Then the difference of the two numbers is :**

A. 7 C. 5
B. 6 D. 8

19. **Simplify** $\sqrt{5 + 2\sqrt{6}} - \frac{1}{\sqrt{5+2\sqrt{6}}} =$

A. $2\sqrt{2}$ C. $1 + \sqrt{5}$
B. $2\sqrt{3}$ D. $\sqrt{5} - 1$

20. $\sqrt{2^4} + \sqrt[3]{64} + \sqrt[4]{2^8} =$

A. 12 C. 18
B. 16 D. 24

21. $2 \times \sqrt[3]{32} - 3\sqrt[3]{4} + \sqrt[3]{500} =$

A. $4\sqrt[3]{6}$ C. $6\sqrt[3]{4}$
B. $\sqrt[3]{24}$ D. 916

22. $8^{\frac{2}{3}} =$

A. 5 ½ C. 4
B. $21\frac{1}{3}$ D. $3\frac{1}{3}$

23. $16^{\frac{3}{4}} =$

A. $4\sqrt{2}$ C. $2\sqrt{2}$
B. 8 D. 16

24. **Simplify** $(16^{\frac{3}{2}} + 16^{\frac{-3}{2}}) =$

A. 0 C. $\frac{4097}{64}$
B. 1 D. $\frac{16}{4097}$

SET 3
BODMAS

1. $\dfrac{3\frac{1}{4} - \frac{4}{5} of \frac{5}{6}}{4\frac{1}{3} \div \frac{1}{5} - [\frac{3}{10} + 21\frac{1}{5}]} =$

A. $\frac{31}{2}$ C. $\frac{1}{4}$
B. $\frac{5}{6}$ D. $\frac{3}{10}$

$\sqrt{\frac{130}{100}} +$ $10\sqrt{13} +$ $\sqrt{\frac{130}{10000}}$ 2 $\Rightarrow 100\sqrt{13}$

2. $\dfrac{4.41 \times 0.16}{2.1 \times 1.6 \times 0.21} =$

A. 1 C. 0.01
B. 0.1 D. 10

3. $\dfrac{[5+5+5+5] \div 5}{3+3+3+3 \div 3} =$

A. 1 C. $\frac{4}{9}$
B. $\frac{3}{10}$ D. $\frac{2}{5}$

4. $\dfrac{\frac{1}{3} + \frac{1}{4}[\frac{2}{5} - \frac{1}{2}]}{1\frac{2}{3} of \frac{3}{4} - \frac{3}{4} of \frac{4}{5}} =$

A. $\frac{37}{78}$ C. $\frac{74}{78}$
B. $\frac{37}{13}$ D. $\frac{74}{13}$

5. $\frac{0.04}{0.03}$ of $\dfrac{(3\frac{1}{3} - 2\frac{1}{2}) \div \frac{1}{2} of 1\frac{1}{4}}{\frac{1}{3} + \frac{1}{5} of \frac{1}{9}} =$

A. 1 C. 5
B. ½ D. $\frac{1}{5}$

6. $\dfrac{0.3555 \times 0.5555 \times 2.025}{0.225 \times 1.7775 \times 0.2222} =$

A. 5.4 C. 4.5
B. 4.58 D. 5.45

7. $100 \times 10 - 100 + 2000 \div 100$

A. 29 C. 980
B. 920 D. 1000

8. $3\frac{3}{5} \times 3\frac{3}{5} + 2 \times 3\frac{3}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{2}{5}$

A. 15 C. 17
B. 16 D. 18

9. $3\frac{1}{2} - [2\frac{1}{4} \div \{1\frac{1}{4} - \frac{1}{2}(1\frac{1}{2} - \frac{1}{3} - \frac{1}{6})\}]$

A. ½ C. 3 ½
B. 2 ½ D. 9 ½

10. $5\frac{1}{3} \div 1\frac{2}{9} \times ¼ \left[10 + \frac{3}{1 - \frac{1}{5}}\right] =$

A. 15 C. $\frac{128}{11}$
B. $\frac{67}{25}$ D. $\frac{128}{99}$

11. **If $x\ [-2\{-4(-a)\}] + 5\ [-2\{-2(-a)\}] = 4a$; find x ?**

A. − 2 C. − 4
B. − 3 D. − 5

12. $3 \div [(8 - 5) \div \{(4 - 2) + (2 + \frac{8}{13})\}] =$

A. $\frac{15}{17}$ C. $\frac{60}{13}$
B. $\frac{13}{60}$ D. $\frac{17}{15}$

13. $\frac{4}{15}$ **of** $\frac{5}{8} \times 6 + 15 - 10 =$

A. 6 C. 5
B. 3 D. 4

14. **Let $\sqrt{13} = 3.605$, $\sqrt{130} = 11.40$. Find $\sqrt{1.3} + \sqrt{1300} + \sqrt{0.013} = ?$**

A. 36.164 C. 37.304
B. 36.304 D. 37.164

15. $\dfrac{[2.644]^2 - [2.356]^2}{0.288} =$

A. 1 C. 5
B. 4 D. 6

16. $\dfrac{[3.4567]^2 - [3.4533]^2}{0.0034} =$

A. 6.91 C. 6.81
B. 7 D. 7.1

17. $\dfrac{[0.03]^2 - [0.01]^2}{0.03 - 0.01} =$

A. 0.02 C. 0.04
B. 0.004 D. 0.4

18. $[\sqrt{72} - \sqrt{18}] \div \sqrt{12} =$

A. $\sqrt{6}$ C. $\frac{\sqrt{2}}{3}$
B. B. $\frac{\sqrt{3}}{2}$ D. $\frac{\sqrt{6}}{2}$

19. $\dfrac{\sqrt{80} - \sqrt{112}}{\sqrt{45} - \sqrt{63}} =$

A. ¾ C. $1\frac{1}{3}$

B. $1\frac{3}{4}$ D. $1\frac{7}{9}$

20. $\sqrt{0.01}+\sqrt{.81}+\sqrt{1.21}+\sqrt{0.0009}=$

A. 2.1 C. 2.03

B. 2.13 D. 2.4

21. $\sqrt{3\frac{33}{64}}\div\sqrt{9\frac{1}{7}}\times 2\sqrt{3\frac{1}{9}}=$

A. $\frac{45}{256}$ C. $4\frac{3}{8}$

B. $1\frac{17}{28}$ D. $2\frac{3}{16}$

22. $\sqrt{\frac{0.49}{0.25}}+\sqrt{\frac{0.81}{0.36}}=$

A. $7\frac{9}{10}$ C. $2\frac{9}{10}$

A. $9\frac{9}{10}$ D. $\frac{9}{10}$

23. Square root of

$\frac{\left(3\frac{1}{4}\right)^4-\left(4\frac{1}{3}\right)^4}{\left(3\frac{1}{4}\right)^2-\left(4\frac{1}{3}\right)^2}=$ → can be done only through system elimination.

A. $7\frac{1}{12}$ C. $1\frac{1}{12}$ → $\sqrt{\frac{169}{16}+\frac{169}{9}}$

B. $5\frac{5}{12}$ D. $1\frac{7}{12}$ ⇒ $\sqrt{\frac{169(16+9)}{16\cdot 9}}$

24. $\sqrt{4+\sqrt{44+\sqrt{10000}}}=$ → $\frac{13\cdot 5}{4\cdot 3}$

A. 12 B. 8 C. 4 D. −4

SET 4

1. $\left(\frac{1}{2}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}\right)\div\left(\frac{2}{5}-\frac{5}{9}+\frac{3}{5}-\frac{7}{18}\right)=$

A. $5\frac{1}{10}$ C. $3\frac{1}{6}$

B. $2\frac{1}{18}$ D. $3\frac{3}{10}$

2. $8\frac{1}{2}$ - $\left[3\frac{1}{4}\div\left\{1\frac{1}{4}-\frac{1}{2}\left(1\frac{1}{2}-\frac{1}{3}-\frac{1}{6}\right)\right\}\right]=$

A. $4\frac{1}{2}$ C. $9\frac{1}{2}$

B. $4\frac{1}{6}$ D. $\frac{2}{9}$

3. $\frac{\frac{5}{3}\times\frac{7}{51}\ of\ \frac{17}{5}-\frac{1}{3}}{\frac{2}{9}\times\frac{5}{7}\ 0f\ \frac{28}{5}-\frac{2}{3}}=$

A. ½ C. 2

B. 4 D. ¼

4. $1\frac{2}{3}\div\frac{2}{7}\times\frac{a}{7}=1\frac{1}{4}\times\frac{2}{3}\div\frac{1}{6}$. **Find a ?**

A. $\frac{1}{6}$ C. 0.006

B. 0.6 D. 6

5. $9-1\frac{2}{9}$ **of** $3\frac{3}{11}\div 5\frac{1}{7}$ **of** $\frac{7}{9}=$

A. 8 C. $8\frac{32}{81}$

B. 9 D. ¾

6. $\frac{5}{1\frac{7}{8}\ of\ 1\frac{1}{3}}\times\frac{2\frac{1}{10}}{3\frac{1}{2}}$ **of** $1\frac{1}{4}=$

A. $1\frac{1}{2}$ C. 1

B. 0.05 D. 2

7. **If** $\frac{1120}{\sqrt{p}}=80$ **, find p ?**

A. 14 C. 196

B. 140 D. 225

8. $\frac{3\frac{1}{4}-\frac{4}{5}\ of\ \frac{5}{6}}{4\frac{1}{3}\div\frac{1}{5}-\left(\frac{3}{10}+21\frac{1}{5}\right)}-\left(1\frac{2}{3}\ of\ 1\frac{1}{2}\right)$

A. 9 C. 13

B. 11 ½ D. 15 ½

9. $1\div[1+1\div\{1+1\div(1+1\div 2)\}]=$

A. 1 B. $\frac{5}{8}$ C. 2 D. ½

10. $\frac{\frac{1}{3}\div\frac{1}{3}\times\frac{1}{3}}{\frac{1}{3}\div\frac{1}{3}\ of\ \frac{1}{3}}-\frac{1}{9}=$

A. 0 B. 1 C. $\frac{1}{3}$ D. $\frac{1}{9}$

11. $\frac{2\frac{3}{4}}{1\frac{5}{6}}\div\frac{7}{8}\times\left(\frac{1}{3}+\frac{1}{4}\right)+\frac{5}{7}\div\frac{3}{4}\ of\ \frac{3}{7}=$

A. $\frac{56}{77}$ C. $\frac{2}{3}$

B. $\frac{49}{80}$ D. $3\frac{2}{9}$

12. $\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\div\frac{4}{7}\left(\frac{2}{5}+\frac{3}{10}\right)\ of\ \frac{\frac{1}{2}+\frac{1}{3}}{\frac{1}{2}-\frac{1}{3}}$

A. $\frac{2}{3}$ C. 37 ½

B. $\frac{3}{2}$ D. $18\frac{3}{8}$

13. $[0.9-\{2.3-3.2-(7.1-5.4-3.5)\}]=$

A. 0.18 C. 0

B. 1.8 D. 2.6

14. $\left(\frac{5}{2}+\frac{3}{2}\right)\left(\frac{25}{4}-\frac{15}{4}+\frac{9}{4}\right)=$

A. 38 B. 37 C. 19 D. 36

15. $(0.2\times 0.2+0.01)[0.1\times 0.1+0.02]^{-1}$

A. $\frac{5}{3}$ C. $\frac{41}{4}$

B. $\frac{41}{12}$ D. $\frac{9}{5}$

16. $\frac{1}{2}+\left\{4\frac{3}{4}-\left(3\frac{1}{6}-2\frac{1}{3}\right)\right\}=$

A. $3\frac{2}{3}$ C. $4\frac{5}{12}$

B. 1 ¼ D. $1\frac{2}{3}$

17. $\frac{1\frac{1}{4}\div 1\frac{1}{2}}{\frac{1}{15}+1-\frac{1}{10}}=$

A. 3 C. $\frac{2}{5}$

B. 6 D. 5 $\frac{25}{29}$

18. $\frac{-\frac{1}{2}-\frac{2}{3}+\frac{4}{5}-\frac{1}{3}+\frac{1}{5}+\frac{3}{4}}{\frac{1}{2}+\frac{2}{3}-\frac{4}{3}+\frac{1}{3}-\frac{1}{5}-\frac{4}{5}}=$

A. $\frac{-10}{3}$ C. $\frac{-3}{10}$

B. 1 D. 2

19. $\frac{17}{15}\times\frac{17}{15}+\frac{2}{15}\times\frac{2}{15}-\frac{17}{15}\times\frac{4}{15}=$

A. 0 C. 10

B. 1 D. 11

20. $\left[4\frac{11}{15}+\frac{15}{71}\right]^2-\left[4\frac{11}{15}-\frac{15}{71}\right]^2$

A. 1 B. 2 C. 3 D. 4

21. $\left[\sqrt{2}+\frac{1}{\sqrt{2}}\right]^2=$

A. 2 ½ C. 4 ½

B. 3 ½ D. 5 ½

22. $\frac{0.1\times 0.1\times 0.1+0.02\times 0.02\times 0.02}{0.2\times 0.2\times 0.2+0.04\times 0.04\times 0.04}$

A. 0.0125 B. 0.25

C. 0.125 D. 0.5

23. $(71\times 29+27\times 15+8\times 4)=$

A. 3450 C. 3458

B. 2496 D. None of these

SET 5

1. **If** $a=7-4\sqrt{3}$**;find** $a^{\frac{1}{2}}+a^{\frac{-1}{2}}$ **?**

A. $3\sqrt{3}$ C. 4

B. 7 D. $3\sqrt{3}$

2. **If** $x=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$, $y=\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ **Then x + y ?**

A. 8 C. $2\left(\sqrt{5}+\sqrt{3}\right)$

B. 16 D. $2\sqrt{15}$

3. **If** $27^{2x-1}=243^3$**, find x ?**

A. 3 C. 7

B. 6 D. 9

4. **If** $3^{x+8}=27^{2x+1}$**, find x ?**

A. 7 C. 2

B. 3 D. 1

5. $36^{\frac{1}{6}}=$

A. 1 C. $\sqrt{6}$

B. 6 D. $\sqrt[3]{6}$

6. $\left[\frac{8}{125}\right]^{\frac{-4}{3}}=$

A. $\frac{625}{16}$ B. $\frac{625}{8}$

C. $\frac{625}{32}$ D. $\frac{16}{625}$

7. **If** $5\sqrt{5}\times 5^3\div 5^{\frac{-3}{2}}=5^{a+2}$**, find a ?**

A. 4 B. 5 C. 6 D. 8

8. **If** $3^{2x-y}=3^{x+y}=\sqrt{27}$ **; find** 3^{x-y} **?**

A. 3 C. $\frac{1}{\sqrt{27}}$

B. $\frac{1}{\sqrt{3}}$ D. $\sqrt{3}$

9. **If** $x=\frac{\sqrt{7}-1}{\sqrt{7}+1}-\frac{\sqrt{7}+1}{\sqrt{7}-1}=a+\sqrt{7}\,b$ **; find a and b ?**

A. $\sqrt{7}$, 1 C. 0, $\frac{-2}{3}$

B. $\sqrt{7}$, -1 D. $\frac{-2}{3}$, 0

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    A. 8.661 C. 1.7321
    B. 4.331 D. -1.732

16. **If $\sqrt{4096}$ =64; find $\sqrt{40.96}+\sqrt{0.4096}+\sqrt{0.004096}+\sqrt{0.00004096}$ =**
    A. 7.09 C. 7.11
    B. 7.10 D. 7.12

17. $$\sqrt{8+\sqrt{57+\sqrt{38+\sqrt{108+\sqrt{169}}}}}$$
    A. 4 B. 6 C. 8 D. 10

18. **If 10.15^2 = 103.02225 , then**
    $$\sqrt{1.030225}+\sqrt{10302.25}$$
    =
    A. 1025.15 C. 103.515
    B. 102.515 D. 102.0515

19. **If $\sqrt{18225}$ = 135; then $\sqrt{18225}+\sqrt{182.25}+\sqrt{1.8225}+\sqrt{0.018225}$**
    A. 14.9985 C. 1499.85
    B. 149.985 D. 1.49985

20. $$\sqrt{21\frac{51}{169}}=$$
    A. $5\frac{8}{13}$ C. $4\frac{3}{13}$
    A. $4\frac{8}{13}$ D. $5\frac{5}{13}$

21. **If $(1101)^2$ = 1212201; find $\sqrt{121.2201}$ =**
    A. 110.01 C. 1.101
    B. 11.01 D. 11.001

22. $$\sqrt{\frac{0.064\times 0.256\times 15.625}{0.025\times 0.625\times 4.096}}=$$
    A. 2 C. 0.24
    B. 2.4 D. 4.2

23. **Find $\sqrt{19.36}+\sqrt{0.1936}+\sqrt{.001936}+\sqrt{0.00001936}$ =**
    A. 4.8484 C. 4.8884
    B. 4.8694 D. 4.8234

24. **Simplify**
    $$\sqrt{3\frac{33}{64}}\div\sqrt{9\frac{1}{7}}\times 2\sqrt{3\frac{1}{9}}$$
    A. $\frac{45}{256}$ C. $4\frac{3}{8}$
    B. $1\frac{17}{28}$ D. $2\frac{3}{16}$

25. **Simplify $\frac{\sqrt{32}+\sqrt{48}}{\sqrt{8}+\sqrt{12}}$ =**
    A. 3 C. 6
    B. 2 D. 4

26. **Simplify**
    $$\frac{3\sqrt{2}}{\sqrt{3}+\sqrt{6}}-\frac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}}+\frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}}$$
    A. $\sqrt{2}$ C. $\sqrt{3}-\sqrt{2}$
    B. $\frac{1}{\sqrt{2}}$ D. 0

27. $\sqrt{\frac{2+\sqrt{3}}{2}}$
    A. $\pm\frac{1}{\sqrt{2}}(\sqrt{3}+1)$
    B. $\pm\frac{1}{2}(\sqrt{3}-2)$
    C. $\pm\frac{1}{2}(\sqrt{3}-1)$
    D. None of these

SET 2

CUBE ROOTS

1. $\frac{\sqrt[3]{8}}{\sqrt{16}}\div\sqrt{\frac{100}{49}}\times\sqrt[3]{125}=$
   A. 7 C. $\frac{7}{100}$
   B. 1 ¾ D. $\frac{4}{7}$

2. $\sqrt[3]{\frac{72.9}{0.4096}}=$
   A. 0.5625 C.182
   B. 5.625 D. 13.6

3. **$(5.5)^3-(4.5)^3$ =**
   A. 1 C. 74.25
   B. 75 D. 75.25

4. $\sqrt[3]{\frac{7}{875}}$ =
   A. $\frac{1}{3}$ C. $\frac{1}{4}$
   B. $\frac{1}{15}$ D. $\frac{1}{5}$

5. $\sqrt[3]{\frac{19}{513}}$ =
   A. $\frac{1}{9}$ C. $\frac{1}{\sqrt{27}}$
   B. $\frac{1}{3}$ D. $\frac{1}{\sqrt{3}}$

6. **If $\sqrt[3]{175616}$ = 56;**